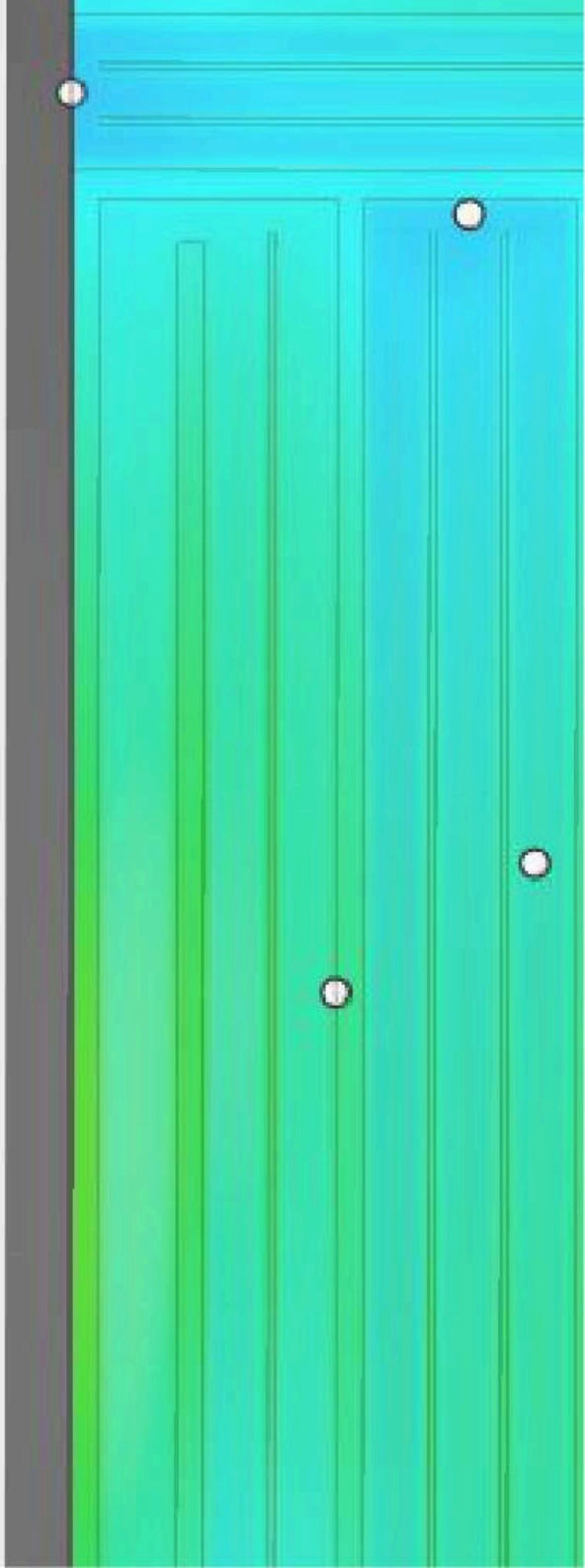
ThermoFluids 2



Charlie Salt, Zayn Muntazir, Zipei Weng, Yousuf Shahabuddin

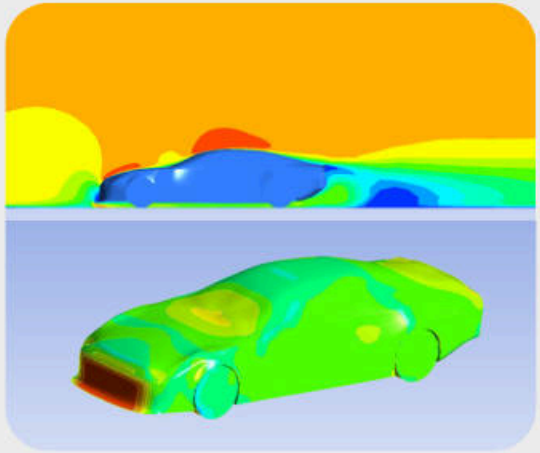


Project Summary

Vehicle Design

A fastback sedan that combines the utility of a sedan with the high-speed capability of a coupe. The design focuses on drag efficiency, controlled rear flow behaviour, high speed stability, simple underbody flow.

The highlighted design features achieve these goals in various ways, such as: the front splitter managing stagnation pressure at the nose and controlling the airflow distribution between the upper body and underbody, the curved roof promoting attached flow and gradual acceleration of air over the car and the rear spoiler acting as a controlled separation device which stabilises the formation of wakes.



Operational Context

Due to their utility and shape, fastback sedans are used by a variety of groups as well at different speeds. In this case summarised as:

Users

- *Daily Commuters*
- *Families*
- *Car Enthusiasts*

Speeds (mph)

Urban	*Rural*	*Motorways*
0-35	35-70	70-100

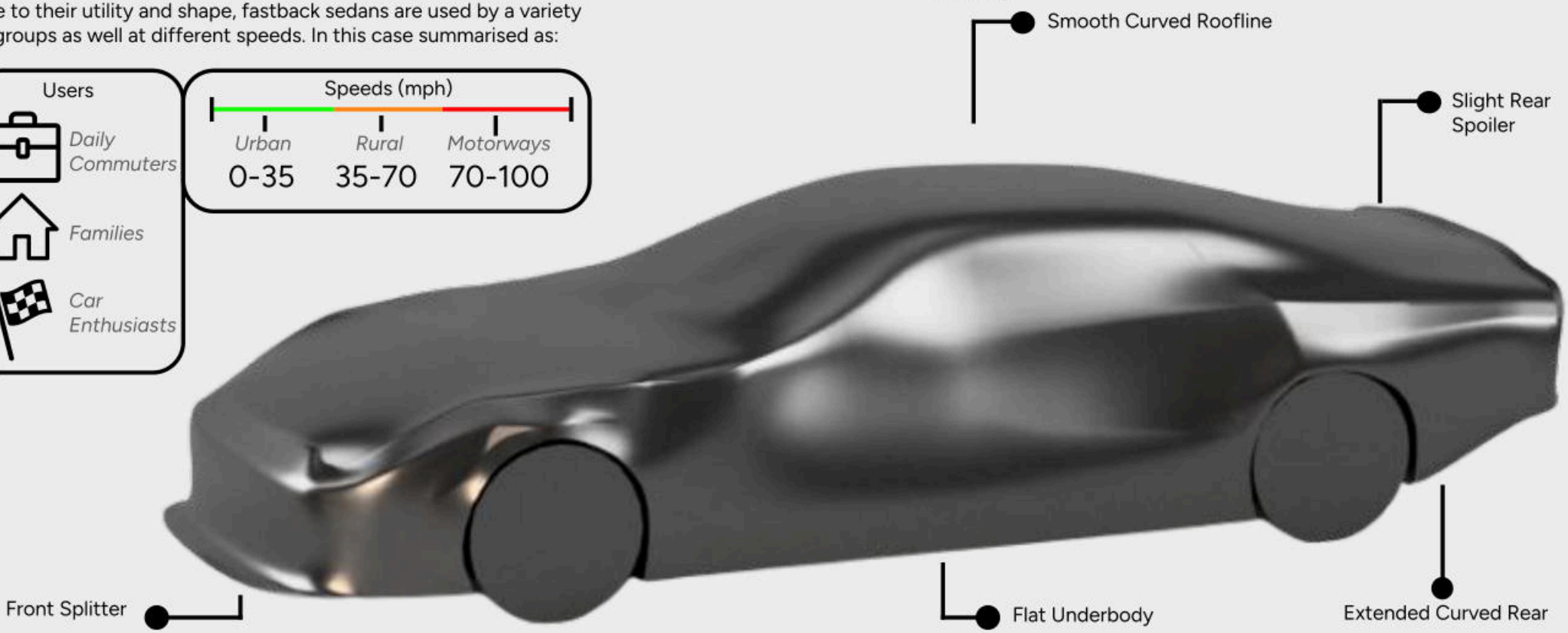
CFD Findings

$$C_d \approx 0.32 \quad C_l \approx -0.05$$

CFD analysis was conducted using ANSYS FluidFlow at a steady-state velocity of 30.5 ms^{-1}. Converging at 200 iterations, the simulation achieved a drag coefficient of 0.32 and a lift coefficient of -0.05, indicating moderate aerodynamic efficiency with slight downforce.

The streamlined roofline and extended rear geometry delayed boundary layer separation, reducing wake size and pressure drag. Smooth pressure gradients and low-pressure regions beneath the vehicle contributed to mild downforce, supported by the flat underbody and front splitter, improving stability.

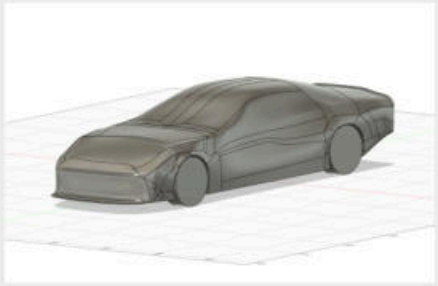
However, rear flow separation and a persistent low-pressure wake remained, limiting drag reduction and indicating incomplete rear pressure recovery.



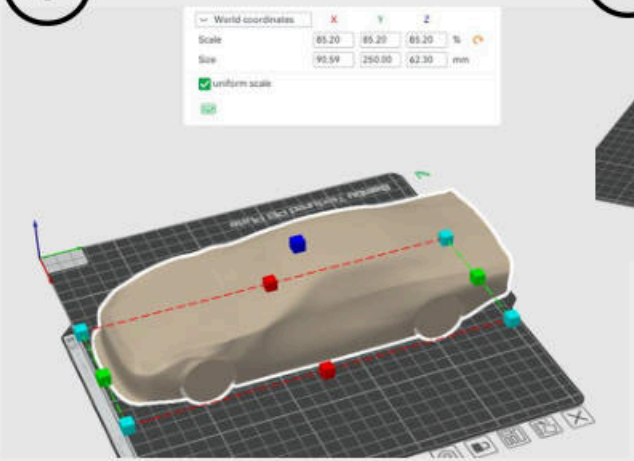
3D Printing

Printing Setup

To validate the CFD analysis previously carried out, wind tunnel testing was underwent. Before that was possible, the vehicle geometry had to be prepared and printed ready for the tunnel.

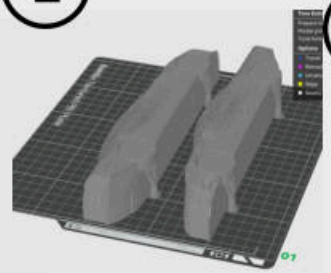


1



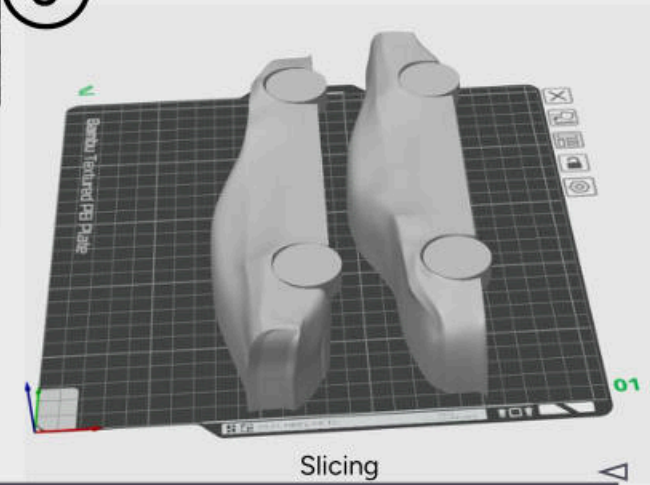
Print Scaling

2



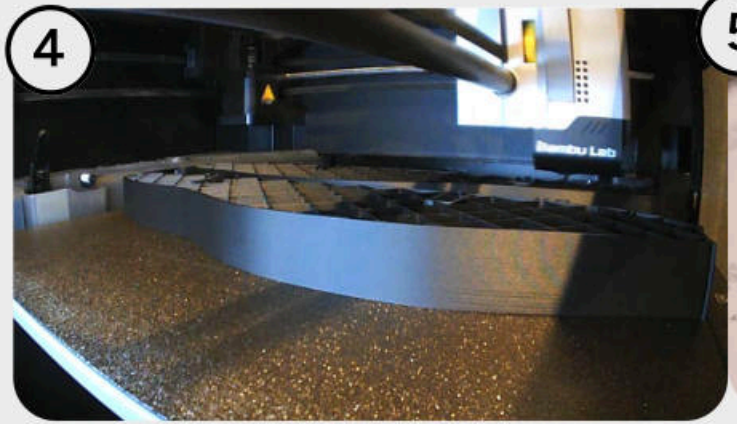
Printing in Halves

3



Slicing

4



Printing

5



Joining Halves

6



Finished Car

3D Print

The dimensions of the CAD required the model to be scaled to fit within the boundaries of the printer we used, in this, scaling down by a factor of 0.85. We were also required to place 4 holes for screws in the underbody. They will be used to secure the car in the wind tunnel.

Assembly + Analysis

We decided to print the car in halves and join the halves. This was because we didn't want the printing lines to have an effect on the values obtained from testing. The choice also came with the added benefits of removing the need for supports and reducing printing time by 25%.

Despite the benefits of our thoughtful decision, it came with the downside of the risk of warping, which became a reality. This meant that when we combined the two halves, the connection wasn't flush. Thankfully, this only causes aesthetic issues, leaving the aerodynamics unchanged as the car is still symmetric, so our results are still trustworthy.

Wind Tunnel Testing

We conducted our wind tunnel testing under conditions that closely matched out CFD simulations of the car, to ensure consistency and accuracy of our results. The test was conducted in a T1, closed-loop wind tunnel to allow for repeatable conditions and therefore reproduceable results. Test conditions: Temperature at 20.9°C, regular air of density 1.225 kg/m³, atmospheric pressure at 101.19 kPa.

As our car was printed at an 85% scale of our CAD, the final print used in the wind tunnel is a 1:20 scale of the full-size car, compared to the 1:18 scale used during our CFD.

Airspeed: Tested at three speeds: 10, 20 and 30.5 ms^{-1}.
Flow visualisation: String Tufts- Flow separation occurs when air can stops following the surface of the object, and so it separates from the surface. This can sometimes happen if there is a sharp change in the geometry of the part. Car manufacturers use string tufts to visualise where this separation occurs, rather than relying purely on calculations- by attaching strings all over the car, you can see where airflow separates as the strings will move around violently. The calmer the string, the smoother the airflow is in that region.
Reynold's number: Calculated from airspeed, car length and air properties:

$$Re = \frac{\rho V L}{\mu} = \frac{(1.225)(31.5)(0.247)}{1.81 \times 10^{-5}} \approx 5.27 \times 10^{5} \quad [1]$$

This value suggest that the flow is mostly turbulent in the wake, while the boundary layer on the body may be partly laminar then transition depending on surface roughness, curvature, and pressure gradients. As it is 3D printed, there is likely to be a high amount of friction at the surface.

Testing Method

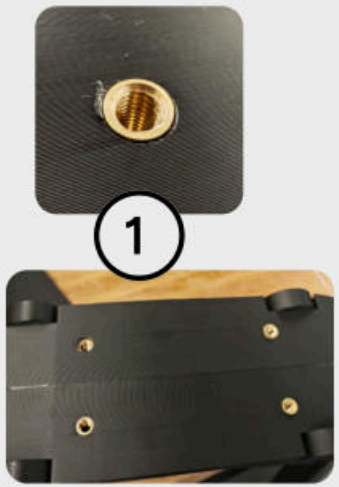
1. We screwed in brass thread inserts, for 100 mm x 40 mm M8 bottoming tapping screws to fix the car securely. Aligned the car on a metal plate and screwed bolts into the car to fix it down securely.

2. We cut some pieces of string for tuft visualisation- using tape, we fixed multiple pieces of string to multiple key points on the car: front, splitter, bonnet, roof, A-pillar, side, and a few points around the rear/ spoiler. This was done to see what parts of the car experienced the most/least turbulent airflow, so we could compare the results with our CFD plots and make better-informed design improvements.

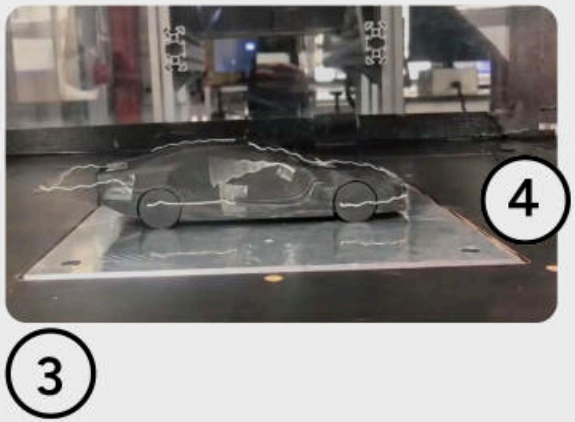
3. Next, we placed the plate with car into the wind tunnel and secured it down with four more bolts.

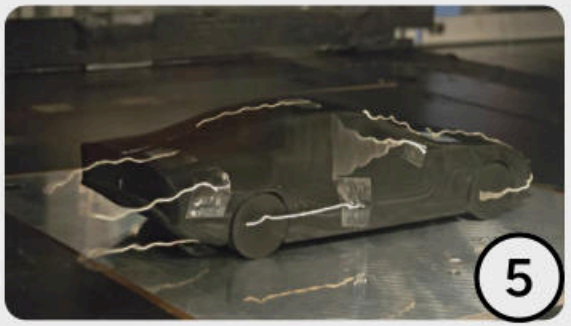
4. The wind tunnel was run at 10 m/s, 20 m/s and 31.5 m/s (CFD simulation ran at 30.5 m/s so our results for 31.5 m/s would be the closest comparison). We did this to evaluate how drag, lift and turbulent flow performs under varying conditions.

5. Results were calculated digitally by the computer, measured every second and compiled into a spreadsheet for us to analyse and edit where necessary.









Analysis

Lift Coefficient plotted against air speed (ms^{-1})



$$C_L = \frac{2F_L}{\rho v^2 A}$$

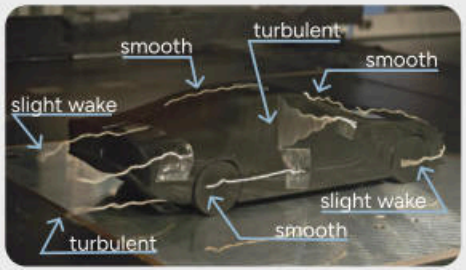
10 ms^{-1}	20 ms^{-1}	30.5 ms^{-1}
Cl: 0.341125	Cl: -1.06295	Cl: -0.38489

Average Cl: -0.3689042

The coefficient of lift value, from our CFD of the car, was $C_L \approx -0.05$, compared to -0.37 from the wind tunnel, suggesting more downforce; although wind tunnels are the most accurate way of measuring this data, there are a few sources for the discrepancies:

The threaded inserts and bolts securing the car may have changed the airflow beneath the car; it is possible they had come out slightly during testing, acting as a sort of diffuser/splitter at the rear/front of the car.

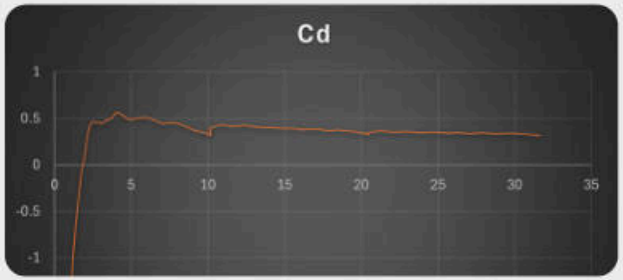
Our CFD mesh resolution only used ~500,000 cells, compared to ~40 million used in industry. The limitation may have missed out on some turbulent flow, especially close to the floor of the car, resulting in the disparity.



The A-Pillar shows turbulent flow, however this is likely due to incorrect tuft placement.

During the testing we estimated the value of the frontal area; we wanted more accurate results, so we recalculated the frontal area to be 0.004632m and performed calculations for the drag and lift coefficients using the rest of the data recorded. The purpose of this analysis was to provide a quantifiable verification of our car's performance. Wind tunnels are the gold standard in industry, and so these results should highlight any discrepancies in our data, or pinpoint areas we may have missed out on during the CFD simulations.

Drag Coefficient plotted against air speed (ms^{-1})



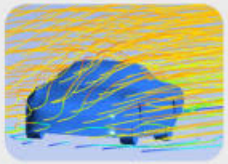
$$C_D = \frac{2F_D}{\rho v^2 A}$$

10 ms^{-1}	20 ms^{-1}	30.5 ms^{-1}
Cd: 0.342391	Cd: 0.331526	Cd: 0.317203

Average Cd: 0.3303734

The coefficient of drag value, from our CFD of the car, was $C_D \approx 0.32$, compared to 0.33 from the wind tunnel results. This is a discrepancy of only 3%, which is great.

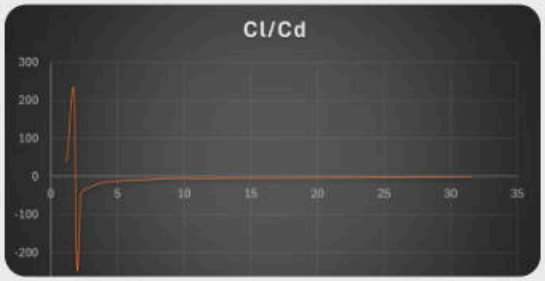
Although small, there is a difference. A likely cause is the 3D printed aspect- these rough lines as a result of the print will have reduced the slipperiness of the body. In the CFD, the car is modelled as a smooth body, and therefore only the shape of the car affected its performance, not the material.



The drag coefficient matches the Porsche Taycan Turbo GT Weissach (0.31), a track-focused road car where 36% of downforce comes from splitters and diffusers. [2]



Drag Coefficient plotted against air speed (ms^{-1})



Frontal Area	Air Speed (m/s)	Lift Force (N)	Drag Force (N)	Side Force (N)	Cd	Cl	Cl/Cd
0.005	31.555	-1.168	0.891	-0.036	0.3154	-0.41346	-1.31089
0.005	31.522	-1.106	0.894	-0.05	0.31713	-0.39233	-1.23714
0.005	31.524	-1.129	0.888	-0.032	0.31496	-0.40044	-1.2714
0.005	31.528	-1.105	0.892	-0.046	0.3163	-0.39183	-1.23879
0.005	31.54	-1.083	0.896	-0.077	0.31748	-0.38373	-1.20871
0.005	31.548	-1.073	0.898	-0.062	0.31802	-0.38	-1.19488
0.005	31.567	-1.103	0.896	-0.069	0.31693	-0.39015	-1.23103
0.005	31.569	-1.091	0.896	-0.041	0.31689	-0.38586	-1.21763
0.005	31.559	-1.023	0.895	-0.067	0.31674	-0.36204	-1.14302
0.005	31.509	-1.102	0.896	-0.053	0.3181	-0.39124	-1.22991
0.005	31.523	-1.067	0.899	-0.064	0.31888	-0.37847	-1.18687
0.005	31.524	-1.049	0.898	-0.066	0.31851	-0.37206	-1.16815
0.005	31.568	-1.045	0.896	-0.072	0.31691	-0.36961	-1.16629
0.005	31.556	-1.121	0.9	-0.065	0.31857	-0.3968	-1.24556
0.005	31.545	-1.045	0.897	-0.076	0.31773	-0.37015	-1.16499
0.005	31.558	-1.083	0.898	-0.064	0.31782	-0.3833	-1.20601
				AVG:	0.31727	-0.38509	-1.21383

Our average Cl/Cd value is -1.21; less negative at higher speeds than lower. It represents "downforce per drag", an efficiency matrix, and as Cl decreases with speed, there is likely some deflection changing pitch with speed. As forces rise, the model could flex a little, reducing its angle of attack. A greater negative number means you get more downforce with the same amount of drag sacrificed, increasing grip. In our case, the car is producing downforce of 1.21 times the car's drag.

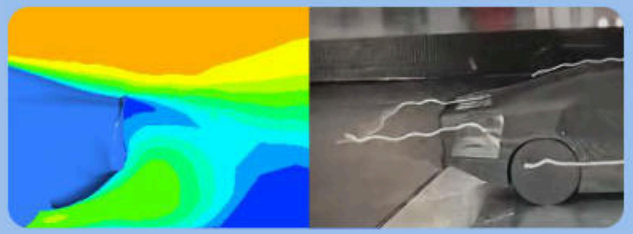
For a road sedan this is not necessarily desirable because it implies extra drag for limited real-world benefit, but a fast-back sedan will benefit from the increased downforce.

Future Recommendations

Based on the comparison between the earlier CFD analysis and the wind tunnel testing, we are able to suggest changes to the design that would improve the aerodynamics of the car for its specific use case.

Issue 1

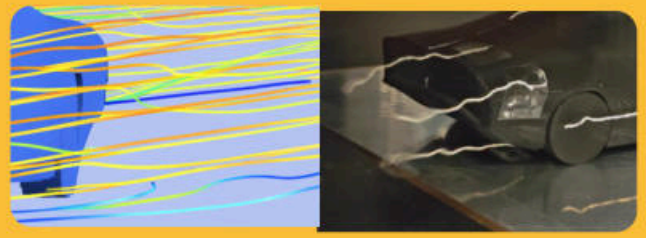
Large wake formation shown by the low pressure region behind the vehicle. This happens when pressure recovery isn't complete. In the best case, we would expect an area of high pressure at the back, shown by a red colour, as we have at the front bumper.



Velocity rear plot | Wind tunnel rear flow

Issue 2

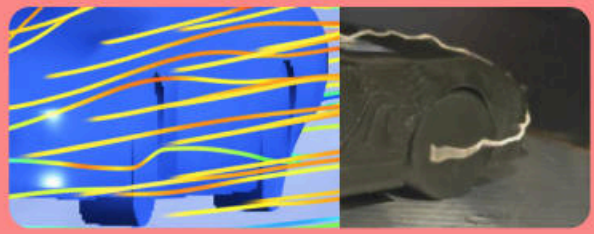
Sligh pressure zone at the rear of the car shown by the green and blue region, with light turbulence indicated by the swaying tufts. The car will get pulled backwards, hence increased drag. We need better pressure recovery and a lower floor pressure.



Velocity rear plot |Wind tunnel diffuser flow

Issue 3

There is a slight wake formation by the front wheel arch and fender area of the car. The body contour is a bit too sharp and does not flow smoothly into the side panels, and is the likely cause for this wake. A slight high pressure region can be seen in the CFD, as well as a turbulent tuft.



Wheel arch airflow CFD | Wind tunnel wheel flow

Change 1

Elongating the tail whilst reducing the slope angle, similar to the Mercedes EQXX, would result in delayed flow separation and improved pressure recovery, overall leading to a smaller wake and lower drag coefficient for the car.



Mercedes EQXX with long tail and tapered roof [3]

Change 2

Adding a rear diffuser will accelerate the underbody flow, increase pressure recovery at the rear and thereby improving downforce. This is expected to reduce wake size and give a more negative lift coefficient, hence greater downforce.



Bugatti Chiron Super Sport diffuser [4]

Change 3

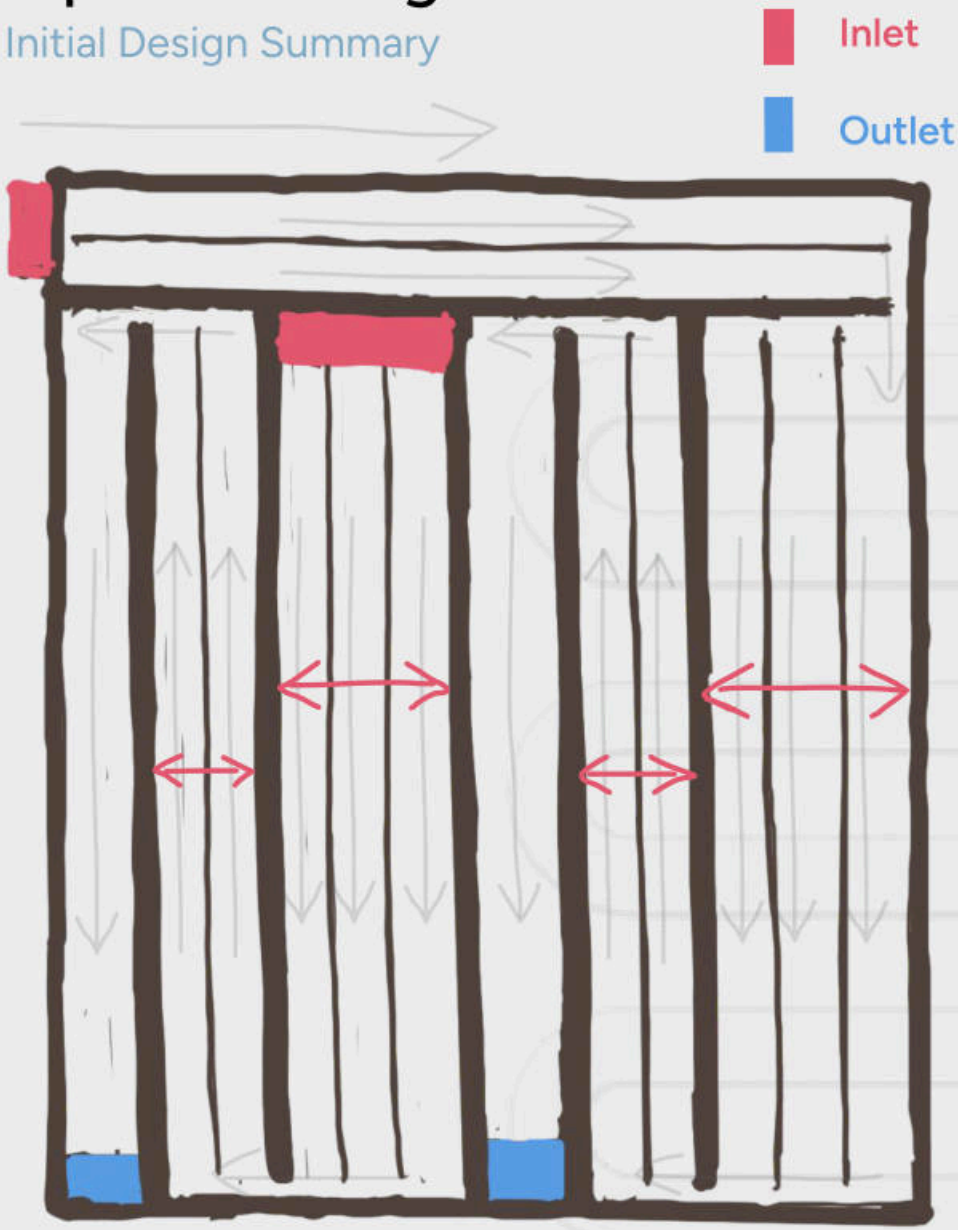
Adding air vents to the frontal sides of the car allows air to flow smoothly over the wheels, and bypass body panel edges, reducing turbulence at the boundary layer and therefore reducing drag.



Audi RS7 aero vents [5]

Pipeline Design

Initial Design Summary



Intended Application :

The cooling plate is intended for liquid-cooled battery applications requiring uniform temperature control and reliable heat removal under high thermal loads, while maintaining low hydraulic resistance for efficient operation.

Design Description :

Hybrid Multi-Port Channel Layout

The final design adopts a hybrid multi-port channel layout with two inlets and two outlets to improve coolant distribution across the plate. By splitting the flow into multiple entry and exit paths, the effective flow length from inlet to outlet is reduced, lowering overall hydraulic resistance. The multi-port configuration also enhances coolant refresh in different regions of the plate, helping to reduce temperature gradients and mitigate local hot spots.

Graded Channel Width

The channel width is progressively varied along the flow direction, with wider channels positioned closer to the inlet and narrower channels towards the outlet. As the coolant travels downstream, its bulk temperature increases and the local temperature driving force decreases; therefore, narrowing downstream channels helps maintain sufficient flow velocity and heat transfer capability. This graded hydraulic resistance strategy promotes more balanced cooling and reduces the likelihood of thermal non-uniformity across the plate.

MPS Channel Schemes

The design is inspired by MPS (multi-pass serpentine) channel schemes, combining the flow guidance characteristics of serpentine layouts with the pressure-reducing benefits of parallel flow splitting. Instead of a single long serpentine path, the channel is subdivided into multiple controlled passes, shortening the effective friction length while maintaining structured coolant routing. This hybrid strategy achieves improved flow uniformity compared to purely parallel layouts, while significantly reducing pressure drop relative to traditional full-length serpentine channels.

Design rationale

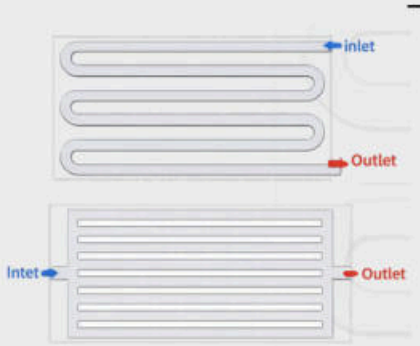
Initial research + Concept Development

Uniform temperature distribution is critical in battery cooling systems. Thermal gradients can accelerate ageing and increase the likelihood of local hot spots.

Tradeoffs:

Flow Path Geometry:

Serpentine and longer paths maintain local flow velocity and more uniform distribution, but increase frictional resistance and overall pressure drop. Parallel paths have the opposite effect.

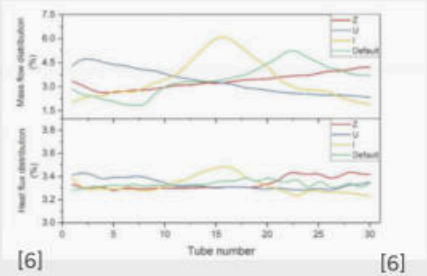


Combine into hybrid layout to balance trade-offs

Inlet/Outlet Configuration

Multiple ports improves the flow distribution and reduces stagnation zones compared to singular ports configurations. The added complexity introduces additional manifolds where

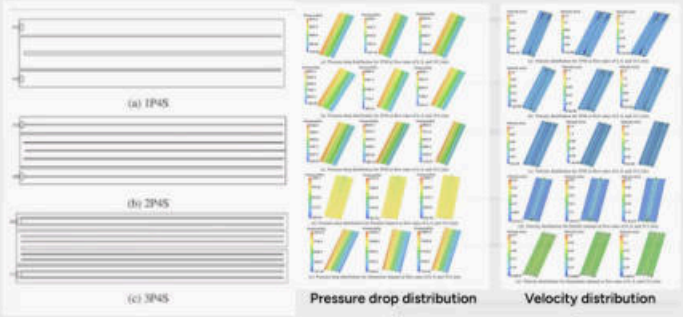




[6] [6]

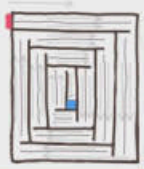
MPS Channel Systems

A hybrid layout that splits flow into several passes, and each pass is a short serpentine section. This keep the uniformity advantage whilst the reducing the pressure drop in the serpentine sections.



Design Iterations

0



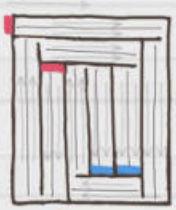
Initial Idea:
Inspired by MPS layouts, multiple parallel passes. Shorter effective path lengths

1



Iteration 1:
Channel widths increased to encourage higher flow distribution towards hotter regions

2

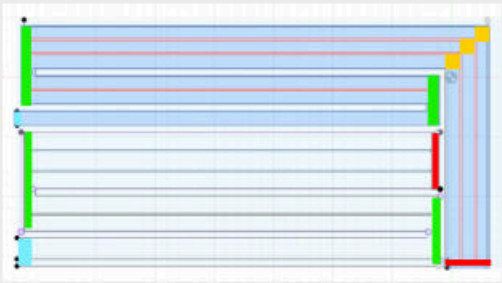


Iteration 2:
To maintain high pressure losses, the design was modified to include dual inlets/outlet. It minimises the travel distance, reducing hydraulic resistance, and improving refresh rate.

Setup for Calculations

The geometry of the coolant channels is necessary to calculate multiple values in both hand calculations. Due to the design, we did not treat the channels as the same size*, although they share the same height of 7mm.

*See exact values for channel geometries in appendix



Benchmarking Values

To gauge the performance of our of our cooling plate, we researched the ideal ranges for a 'mini channel' cooling plate with a hydraulic diameter of 1-10mm.

Name	Benchmarked Value	
Reynolds number	~100–2000	Re
Pressure drop	~20–300	Pa
Max Battery Temperature	39.9	°C
Battery Temp. difference	5	ΔT°C
Cooling Plate Temp. difference	5.4	ΔT°C

Hand Calculation

The following pages cover the hand calculation for the overall pressure drop and heat transfer of the battery to coolant.

Pressure Loss Assumptions

↳ Flow

- Separate pumps for each loop, each pressure dop (1 and 2) are treated as belonging to and independent hydraulic loop
- The coolant flow is assumed to be steady, fully developed, laminar, and incompressible, with constant fluid properties.
- Flow in parallel channels is assumed to split equally; mass flow rate of channel = mass flow rate/ amount of channels its being split into
- Parallel pressure losses are taken as averaged across all parallel channels that the flow splits into (split into 3 channels = the pressure loss is averaged from those three)

↳ Channels

- Channels are rectangular (use the Darcy Weisbach Friction Factor to approximate)
- Non channel sections represented minor loss coefficients, not the exact geometry.
- Surface roughness is neglected
- Slip wall effect effect are included in the friction factor equation not solved directly

Pressure Loss Calculation

To calculate the total pressure loss of both systems, the losses were calculated over every change in geometry to the piping, and then added carefully together at the end (accounting for parallel piping). This is assuming that the each path can be calculated independently.

Flow Properties

1 Divide the mass flow rate by the amount of split channels

2 $A = wh$ Find cross-sectional area

3 $D_h = \frac{2wh}{w+h}$ Find the hydraulic diameter

4 $V = \frac{\dot{m}}{\rho A}$ Find the velocity

5 $Re = \frac{\rho V D_h}{\mu}$ Find the Reynolds Number. All sections were <2000 (laminar)

6 $f = \frac{64}{Re}$ Find the Darcy friction factor

Channel Losses

For each straight channel:

$$\Delta P_{\text{channel}} = f \frac{L}{D_h} \left(\frac{\rho V^2}{2} \right)$$

This calculates the major pressure losses of the system, using the flow properties and the dynamic pressure term.

Minor Losses

For each bend/split/merge/inlet/outlet:

$$\Delta P_{\text{minor}} = K \left(\frac{\rho V^2}{2} \right)$$

This calculates the minor pressure losses of the system, using the loss coefficients (K).

Channel Merge K = 1 | *Inlet loss K = 0.5* | *180° bend K = 2* | *Channel Split K = 0.75*

90° bend K = 1.3 | *Outlet loss K = 1*

Total Pressure Drop

$$\Delta P_{\text{avg}} = \frac{1}{N} \sum_{i=1}^{N} \Delta P_i$$

The average pressure drop was found across each set of parallel channels using this equation.

$$\Delta P_{\text{total}} = \sum_{i=1}^{n} \Delta P_i$$

With the average values for the channels and all minor losses, the total pressure drop was found.

Results

Pa ↓	Loop 1
= 58.03	

Pa ↓	Loop 2
= 48.39	

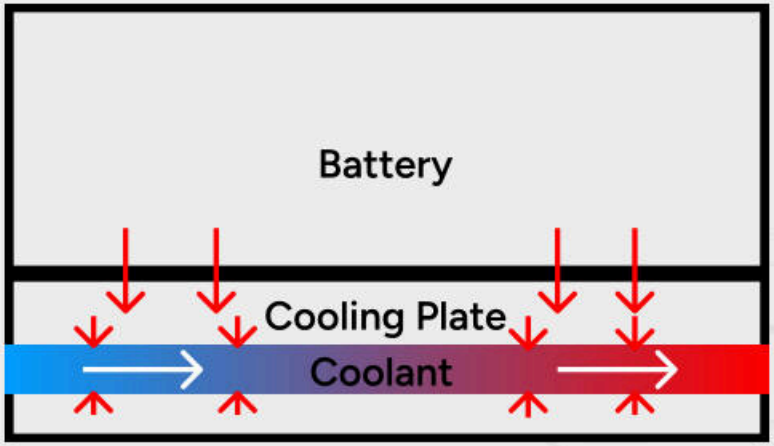
Loop 1 Largest Loss is the outlet channel at 15.26



Loop 2 Largest Loss is the 180 bend at 15.02

Hand Calculation

Heat Transfer Calculation



Assumptions

↳ Battery + Cooling Plate

- Heat flow is one dimensional and is represented as a thermal resistance network(battery to plate (conduction) to coolant (convection)
- Heat is generated uniformly throughout the battery.
- Heat transfer to surrounding area is ignored due to negligible effect
- Plate is homogeneous, the temperature is uniform and the thermal conductivity is constant
- Hydraulic diameter assumed by width

↳ Flow

- There is fully developed flow in the pipe and uniform heat transfer along smooth channel walls
- Assumed laminar flow
- Mass flow rate is assumed through all channel sections
- The two cooling are treated as two parallel heat transfer paths so heat can be transferred into either loop easily
- Assumed linear temperature increase along the cooling paths.
- Assume square channels
- Flow rate splits evenly when the flow separates into multiple channels or merges back

Heat Generation

Battery Volume

$$V = L \times W \times H$$

→ Total Heat Generation

$$Q = q'''V$$

Finding the Nusselt Number

Reynolds Number

$$Re = \frac{\rho v D_h}{\mu}$$

Prandtl Number

$$Pr = \frac{c_p \mu}{k}$$

Nusselt Number

$$Nu = 4.36$$

Previously found all flow is laminar with Reynolds Number. Therefore, Prandtl number is not needed, Nusselt Number is 4.36.

Thermal Resistances

Convective Heat Transfer Coefficient

$$h = \frac{Nu\,k}{D_h}$$

Found the h for all channels, then found the average to use for R_conv equation

Convective Thermal Resistance

$$R_{conv} = \frac{1}{hA_{wet}}$$

L represents the thickness of the cooling plate through which heat conducts from the battery to the coolant channels

K is the thermal conductivity of the material. How easily the heat flows through the material

Conductive Thermal Resistance

$$R_{cond} = \frac{L}{kA}$$

Total Thermal Resistance of One Cooling Path

$$R_{loop} = R_{cond} + R_{conv}$$

Loop 1 Temp:
Average= 27.89
Outlet= 23.94

Loop 2 Temp:
Average= 27.34
Outlet= 23.67

→ **Parallel Combination of Two Thermal Resistances**

$$R_{total} = \left(\frac{1}{R_1} + \frac{1}{R_2}\right)^{-1}$$

Temperature Differences

Battery Temperature Rise Above Coolant

$$\Delta T = QR_{total}$$

Battery Temperature

$$T_{battery} = T_{avg,total} + \Delta T$$

°C ↑	Battery
= 35.25	

Coolant Temperature Rise

$$\Delta T_{coolant} = \frac{Qw}{\dot{m}c_p}$$

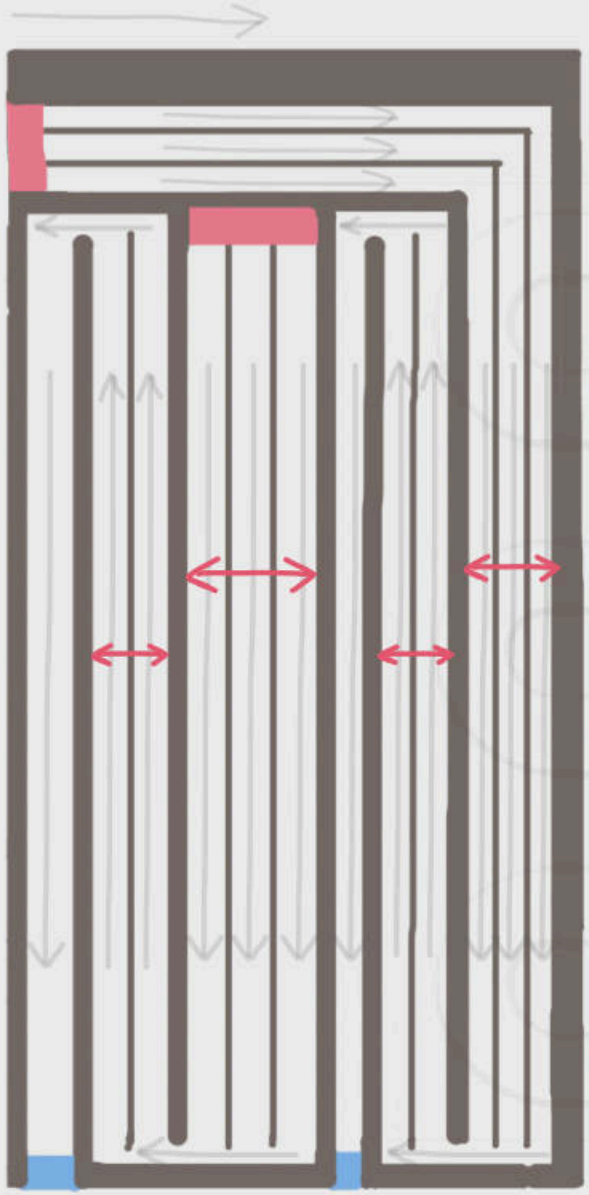
Overall Average Coolant Temperature

$$T_{avg,total} = \frac{T_{avg,1} + T_{avg,2}}{2}$$

°C ↑	Coolant
= 23.81	

3D Modelling

Design choices and Modelling details



Top channels design：

The top channel is split into three branches that connect to the three right-side channels. This modification helps reduce flow deviation and improves coolant distribution across the right-side channels.

Rectangular channels：

Rectangular internal channels are used to increase the heat transfer surface area between the coolant and the cooling plate. This geometry also promotes more uniform flow distribution and is easier to manufacture using common machining processes.

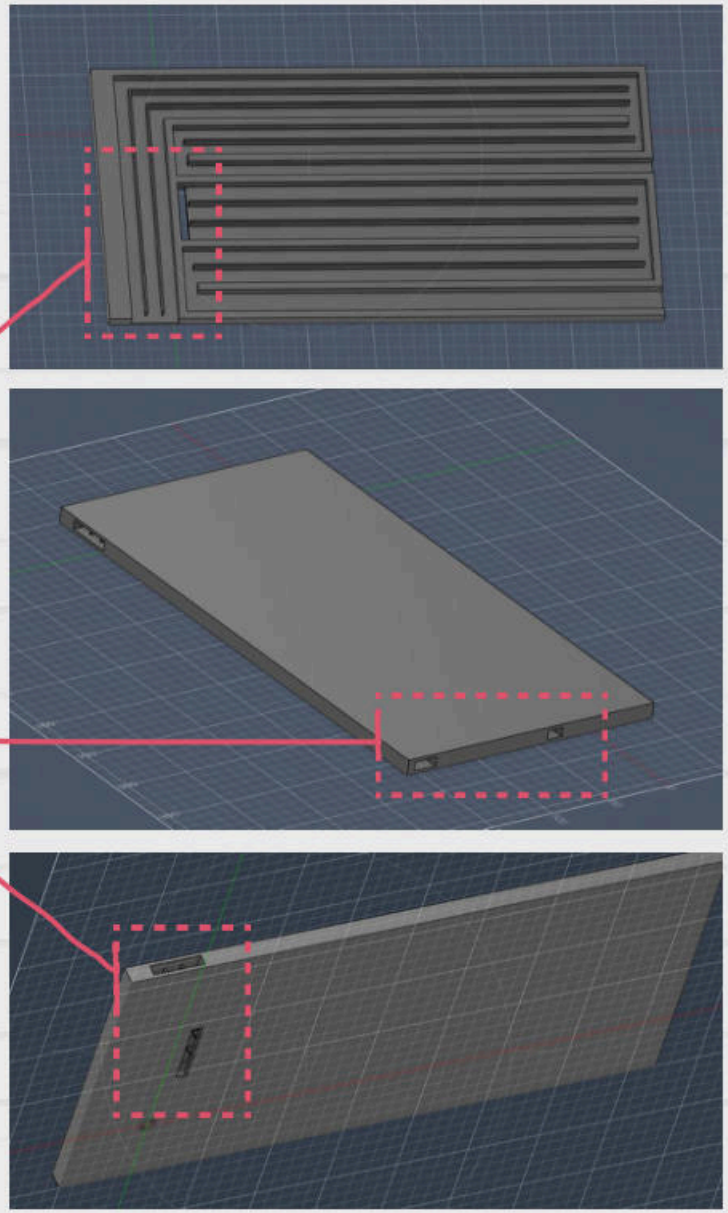
Multiple inlets/outlets：

Two inlets and two outlets are introduced to improve the coolant distribution across the cooling plate. Compared with a single inlet configuration, the dual-inlet design reduces flow path length and helps balance the flow between channels, which can lower pressure drop and reduce the risk of local hotspots.

Inlet

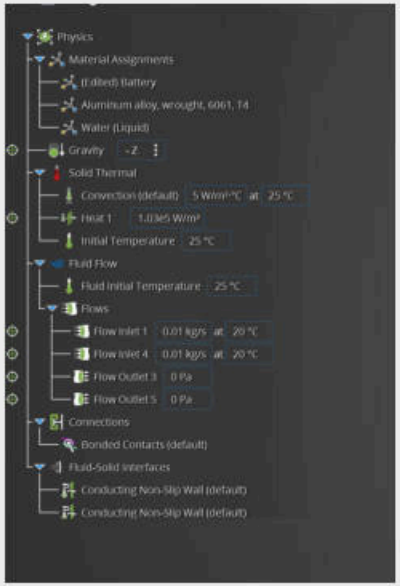
Outlet

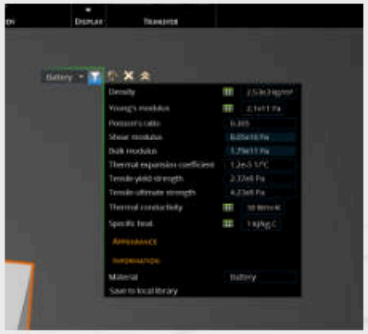
Modeling Details

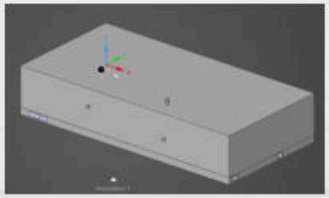


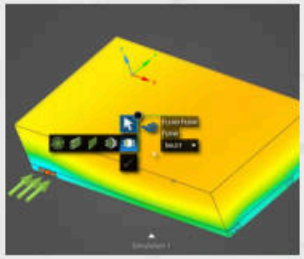
CFD Setup

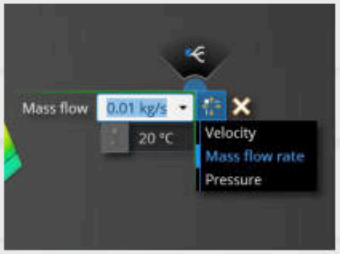
Setup and Assumptions

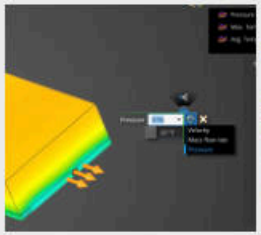


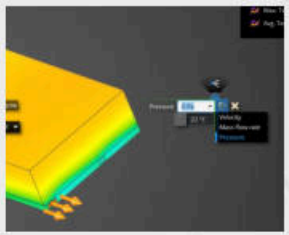


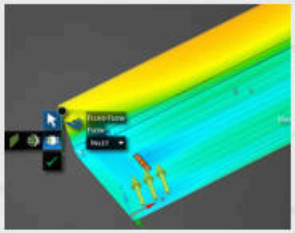


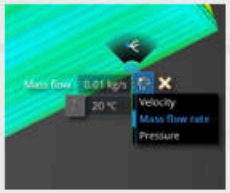


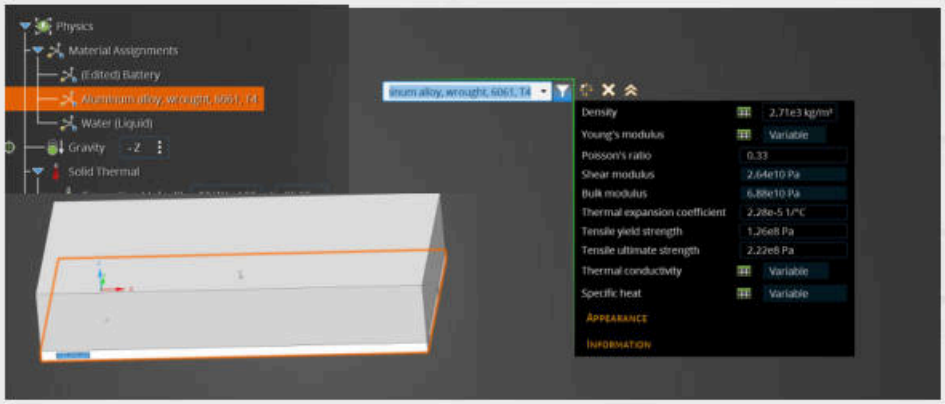












Assumptions

Geometry and modelling :

Battery model: Battery module simplified as a single homogeneous solid block, neglecting internal cell structures

Heat source specification :

Heat generation: Uniform volumetric heat source of 103113 W/m³, representing heat produced during 2C discharge

Fluid properties :

Coolant: Liquid water with constant thermophysical properties.
Flow assumption: Coolant flow treated as incompressible

Simulation setup :

Simulation type: Steady-state simulation
Initial temperature: Battery, coolant and cooling plate initialised at 25°C

Inlet condition :

Two mass-flow inlets: 0.01 kg/s each, inlet temperature 20°C

Outlet condition :

Two pressure outlets: Fixed at 0 Pa (gauge pressure)

Wall condition :

No-slip condition applied at all fluid–solid interfaces. Heat transfer between coolant and plate solved using conjugate heat transfer

External boundary condition :

Ambient convection applied: heat transfer coefficient 5 W/m²·K at 25°C

Material properties :

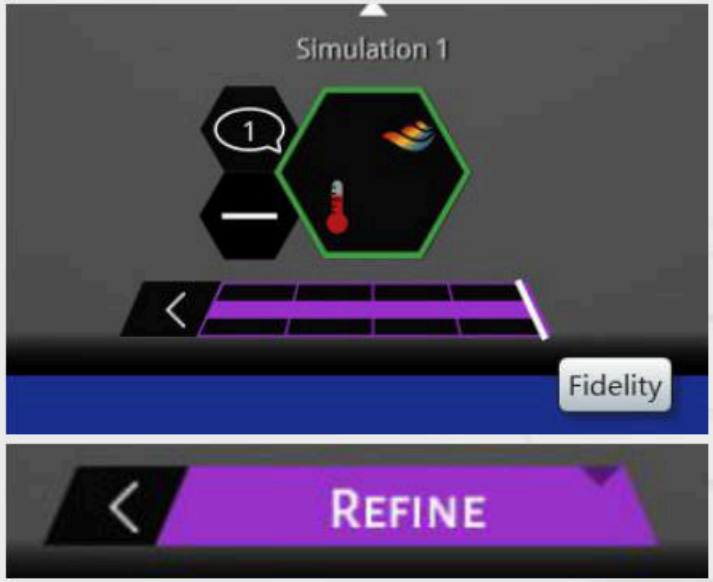
Battery properties: ρ = 2530 kg/m³, k = 30 W/m·K, cp = 1000 J/kg·K
Cooling plate: Aluminium alloy 6061-T4

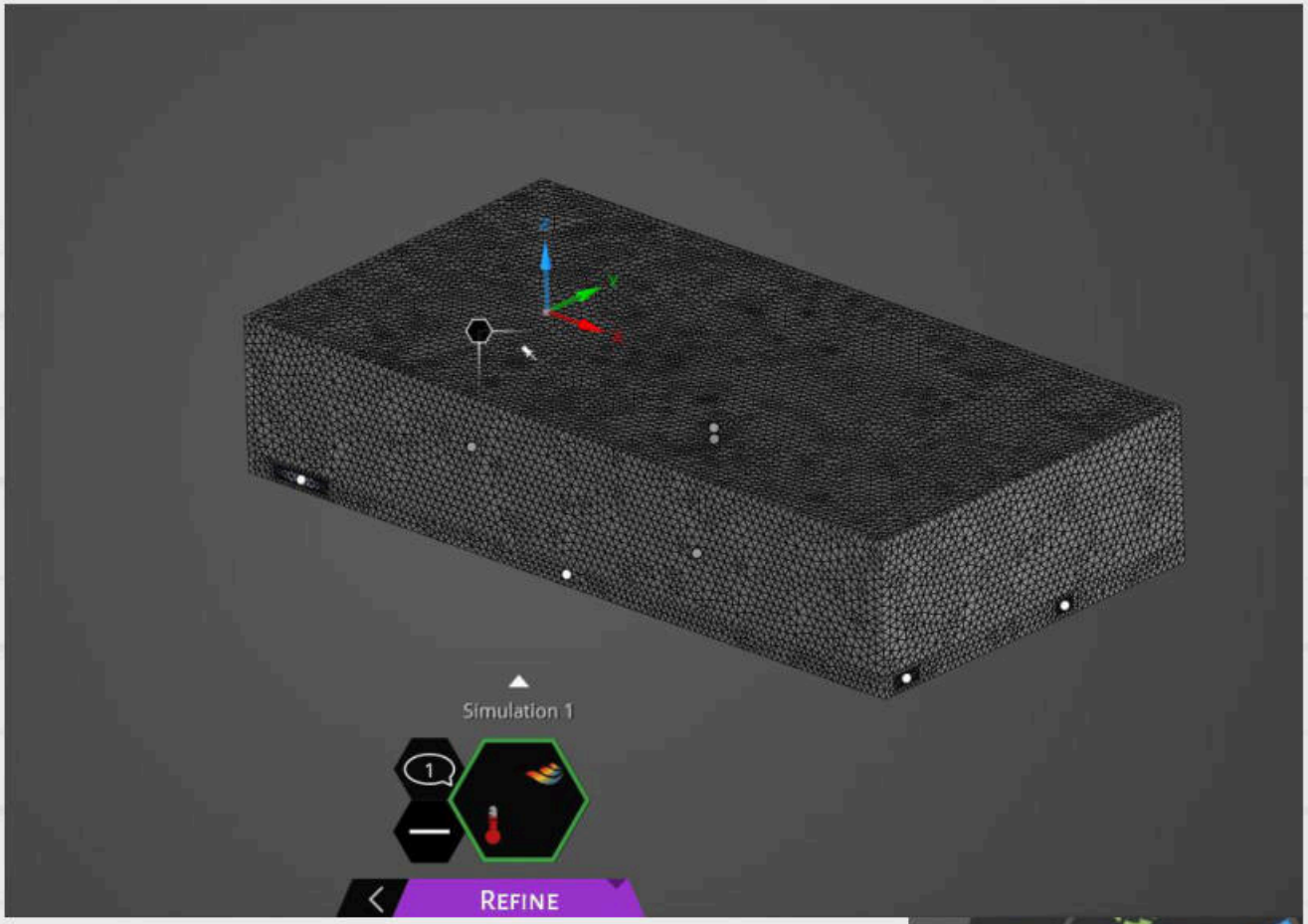
Thermal contact assumption :

Contact thermal resistance between battery and cooling plate is neglected

Meshing

Meshing Strategy





Mesh resolution :

Mesh density controlled using the Refine mode with the highest fidelity level to improve solution accuracy

Mesh generation :

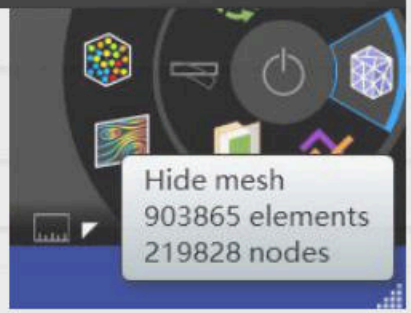
The computational domain was discretised using an automatic mesh generated in ANSYS Discovery

Mesh size :

Total mesh elements: 903,865 cells
Total nodes: 219,828

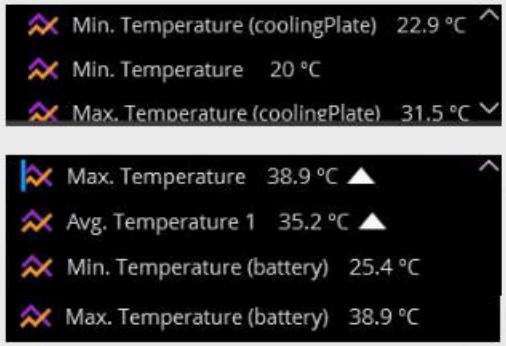
Refinement regions :

Finer mesh automatically generated in cooling channels, where strong velocity and temperature gradients occur. Additional refinement captured flow behaviour near bends and inlet/outlet regions



CFD Outputs

Temperatures and Heat Flux

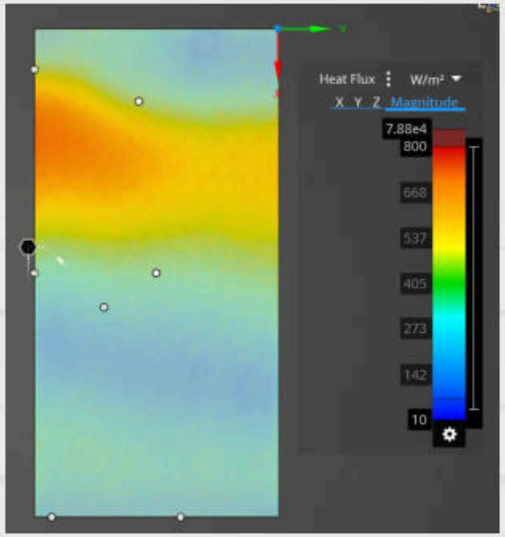


CFD Outputs (Temperatures)

Temperature distribution analysis

The cooling plate temperature map shows a smooth gradient from the inlet toward downstream channels, indicating that coolant is distributed across most of the channel network rather than concentrated in a single path. Most of the cooling plate remains within a relatively narrow temperature range, suggesting that the multi-channel layout promotes fairly uniform heat removal across the battery surface. However, slightly higher temperatures appear in downstream regions, indicating gradual thermal accumulation as coolant temperature increases along the flow path.

The battery temperature distribution further confirms that large localized hot spots are effectively suppressed. The maximum battery temperature remains below 39 °C, while the minimum is around 25 °C, demonstrating that the cooling plate maintains an acceptable thermal gradient across the module. Nevertheless, edge regions closer to the outlet exhibit slightly elevated temperatures, suggesting that coolant effectiveness decreases as it travels downstream.



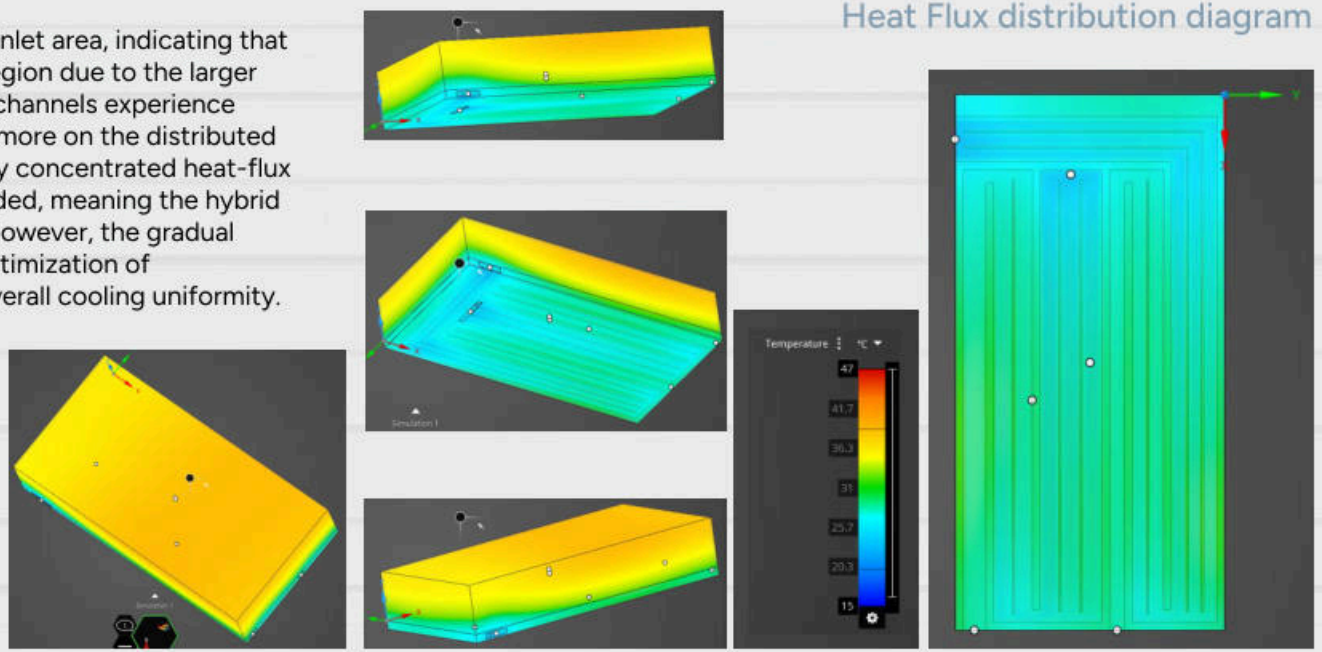
Heat Flux distribution diagram

Local flow behaviour and heat flux

The heat-flux map reveals localized high-flux regions near the inlet area, indicating that the incoming coolant absorbs heat more aggressively in this region due to the larger temperature difference. This also suggests that the upstream channels experience stronger cooling performance, while downstream regions rely more on the distributed channel network to maintain cooling. The absence of extremely concentrated heat-flux zones suggests that severe flow maldistribution is largely avoided, meaning the hybrid multi-channel layout successfully mitigates strong bias flow. However, the gradual reduction of heat flux toward the outlet implies that further optimization of downstream channel width or flow balancing could improve overall cooling uniformity.

Design evaluation

Overall, the CFD results suggest that the proposed cooling plate design performs effectively in reducing hot spots and maintaining relatively uniform temperature distribution across the battery module. The multi-port channel configuration appears to successfully distribute coolant over the cooling plate surface, while the gradual temperature rise downstream highlights a potential opportunity for further optimization of flow balance.



Temperature distribution diagram

CFD Outputs

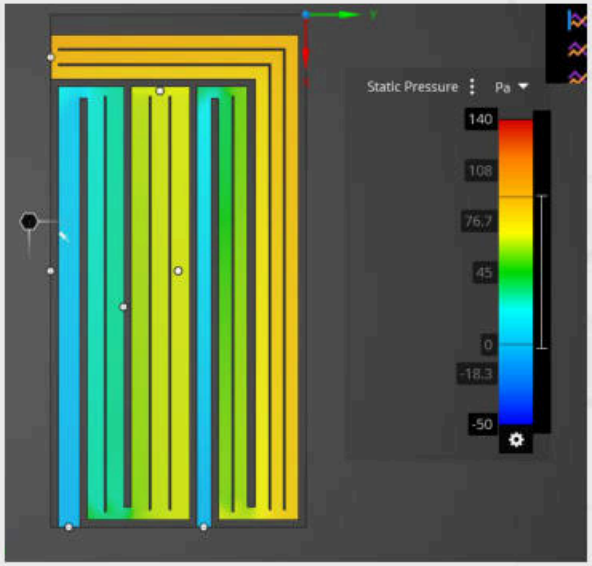
Pressures and Velocity

Pressure and velocity distribution analysis

The velocity plots show that the coolant generally follows the designed channel network and flows through most of the parallel channels across the cooling plate. This indicates that the channel layout successfully distributes coolant across the plate rather than allowing the flow to concentrate in a single dominant path, and no obvious stagnant zones are observed.



CFD Outputs (Pressures and Velocity)



CFD Outputs (Pressures and Velocity)

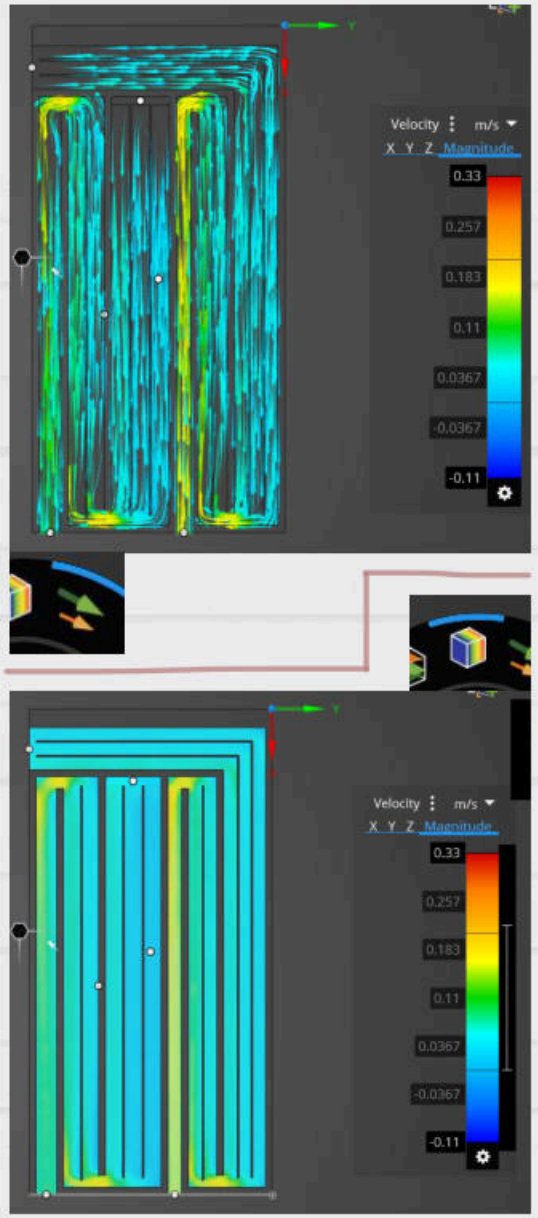
However, the velocity contours also reveal noticeable variations between channels. Some channels exhibit slightly higher velocities, particularly near the inlet manifold and channel turning regions, where the coolant accelerates as it enters and redistributes into the channel network. In contrast, some outer and downstream channels show relatively lower velocities, indicating that these channels receive slightly less flow.

This behaviour reflects flow maldistribution caused by differences in hydraulic resistance between channels. Channels that provide slightly shorter or less resistive paths receive larger flow rates and therefore higher velocities, while channels with longer paths or additional turning losses receive less flow. Although these variations are not severe enough to create stagnant zones, they indicate that the coolant distribution is not perfectly uniform.

The pressure map supports this interpretation. Pressure gradually decreases from the inlet to the outlet, which is consistent with expected frictional losses along the flow path. The overall pressure drop of approximately 69.5 Pa indicates relatively low hydraulic resistance, suggesting that the design maintains good hydraulic efficiency without requiring excessive pumping power.

These flow characteristics also explain the temperature results observed previously. Regions with higher velocities experience stronger convective heat transfer, resulting in more effective cooling, while regions with lower velocities provide weaker heat removal. This explains why the temperature distribution shows slightly higher temperatures in some downstream or edge regions even though severe hot spots are avoided.

Velocity distribution diagrams



Conclusion

Overall, the results suggest that the cooling plate design is successful in distributing coolant across the surface and preventing strong stagnation, while maintaining a relatively low pressure drop. However, the remaining velocity differences between channels indicate that flow balancing is not fully achieved, which leads to the mild temperature gradients observed in the thermal results.

CFD vs Hand Calculations

Pressure comparison:
Treating each cooling loop independentley, the hand calculations get a pressure drop of 58Pa for loop 1 and 131Pa for loop 2. The significant difference between the 2 loops is due to the different path lengths and number of bends.

Sitting between the 2 hand calculations, the CFD simulation computes a value of 69.5Pa, which is noticabely closer to the value for loop 1, which can be explained by many factors:

The hand calculation assumes that the flow splits evenly at each junction, however, in reality, the flow takes the path of least hydraulic resistance, as shown by the CFD. Channels with shorter path lengths and less bends recieve more flow meaning that the pressure drop across the plate is weighted to the lower resistance paths, causing the CFD value to be closer to that of the hand calucucation of loop 1.

As well as this, the hand calculations use fixed minor loss coefficients (K=1.3 for 90 and K=2 for 180). Although these are the standard values, in practice, the values would differ based on the specific geometry. Since the CFD simulation calculates the actual flow through the pipes, it can show the true separation and pressure recovery at each junction.

Despite the differences, the values of the CFD and hand calculations being a similar magnitude suggests that the hand calculations capture the general idea of the physics well.

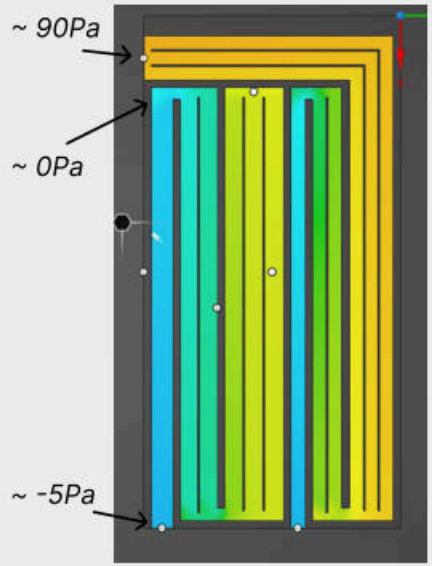
Temperature comparison:
The temperature hand calculation predicts a battery temperature of 35.2C using a 1D thermal resistor network and treating the 2 cooling loops as thermal resistors in parralel. The average coolant temeperature is calculated as 23.8C.

The CFD reports an average battery temperature of 35.2C and a max temperature of 38.9C, with the cooling plate ranging from 22.9C and 31.5C. The average temperature of both methods is the same, which is a strong validation of the thermal resistance approach for estimating bulk performance.

A main difference between them is that the CFD reports specific spatial information that the hand calculations can't. The 38.9°C maximum battery temperature occurs in downstream and edge regions where coolant velocity is lower and the local temperature difference driving heat transfer has decreased. By assuming uniform heat removal across all channel surfaces and a linear coolant temperature rise, the hand calcuation effectively returns a spatially averaged result. It correctly predicts the mean but can't show the roughly 3.7°C spread between average and peak battery temperature that the CFD captures.

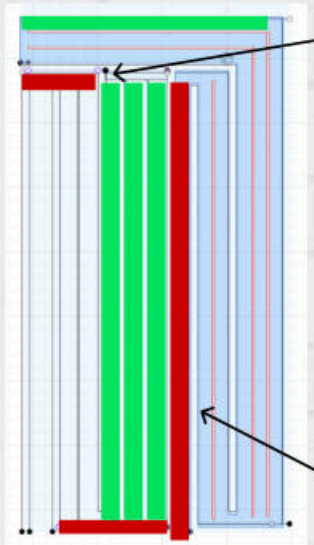
The CFD also applies an ambient convection boundary condition (5 W/m²·K at 25°C) on external surfaces, which the hand calculation neglects. At these values, the ambient heat loss is small compared to the coolant-side heat removal, so its effect on the overall energy balance is minor, but it does contribute slightly to the lower minimum temperatures seen on the cooling plate in the CFD (22.9°C vs. the hand calculation's average coolant temperature of 23.8°C).

CFD

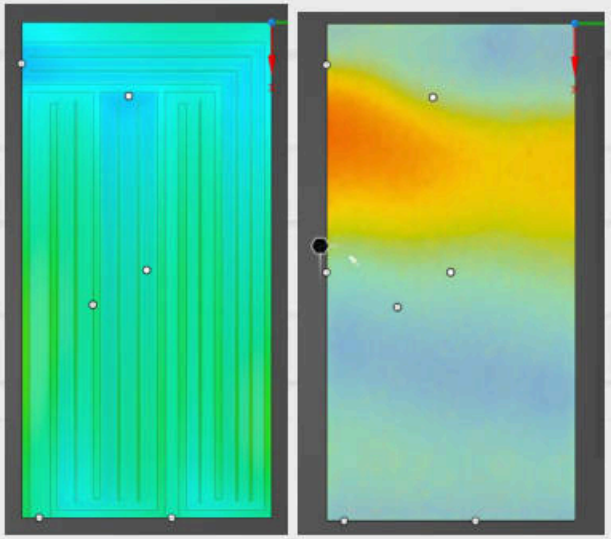


VS

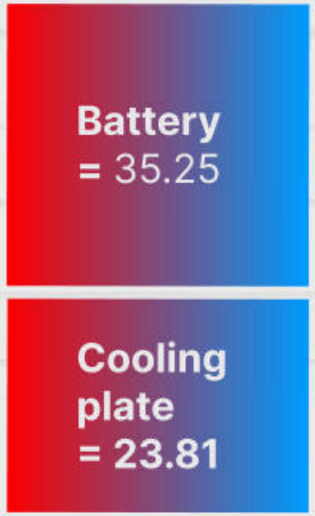
Hand Calculation



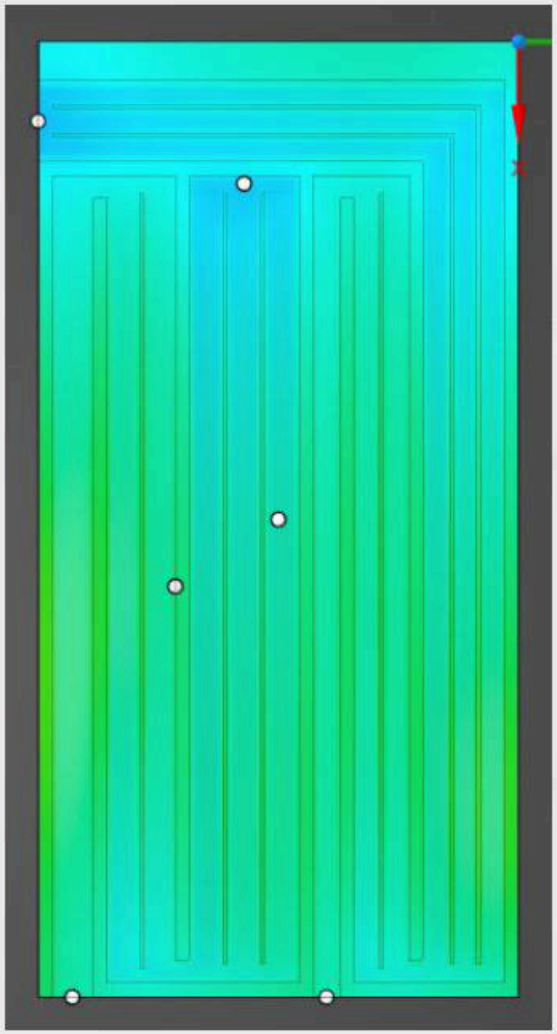
CFD

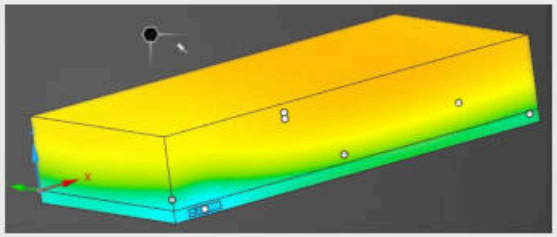


Hand Calculation



Analysis





Results

THERMAL UNIFORMITY

ΔT°C	Temperature Spread (Battery)	ΔT°C	Temperature Spread (Cooling Plate)
13.5		8.6	
↑ Above Limit (5)		↑ Above Limit (5.4)	

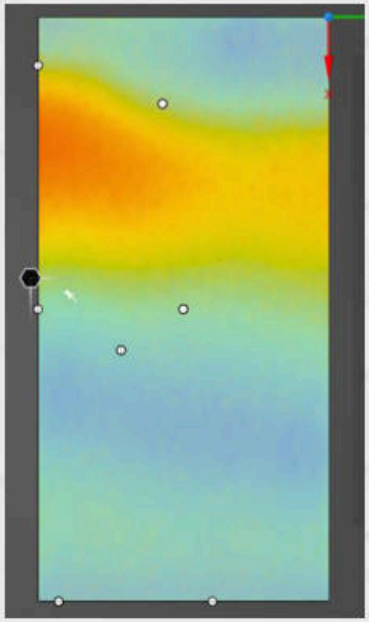
°C ↑ 38.9

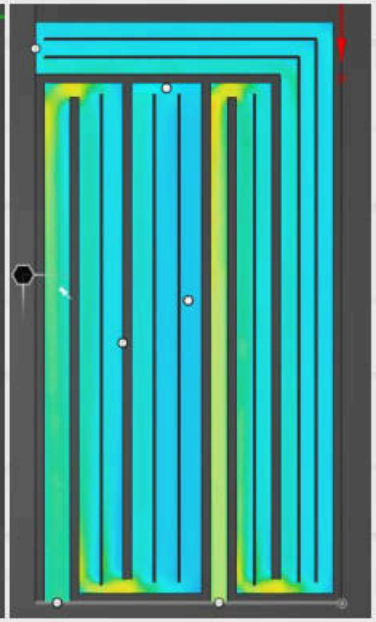
↓Below Limit (39.9)

THERMAL SAFETY

Pa ↓ 69.5 **Pressure Drop**

20 — We are here — 300





Brief Discussion

We designed around achieving balanced coolant distribution and uniform heat transfer. Many design choices such as square channels and varying channel geometries, were expected to introduce additional pressure losses. However, the results show the opposite behaviour. The pressure drop remained low relative to our benchmarked range (20-300Pa)

Thermal Performance

The thermal performance was the focus of the project, and the results show that the system performs effectively in maintaining safe operating temperatures. The maximum temperature of the battery reached 38.9**°C,** below the threshold of 39.9°C, meaning no region of the battery exceeds the safe operating limit. Above this threshold, the higher temperatures would cause the increased electrolyte decomposition, rapidly degrading the battery.

Hydraulic Performance

The velocity contours indicate that the outlet design helps distribute the coolant through the channels, allowing the flow to spread across the plate. This distributed flow helps maintain a generally low temperature across the system, while maintaining a pressure drop of 69.5 Pa.

Uniformity Limitation

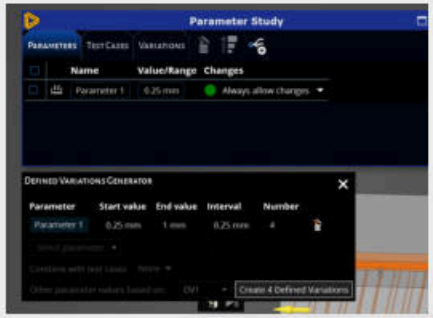
The main limitation of the design is the uniformity of the temperature distribution. The peak battery temperature remains within safe limits, but the temperature spread of 13.5 °C indicates that cooling is not perfectly uniform across the plate. Variations in channel path length and flow velocity mean that some regions remove heat more effectively than others, leading to small but noticeable temperature gradients across the battery surface.

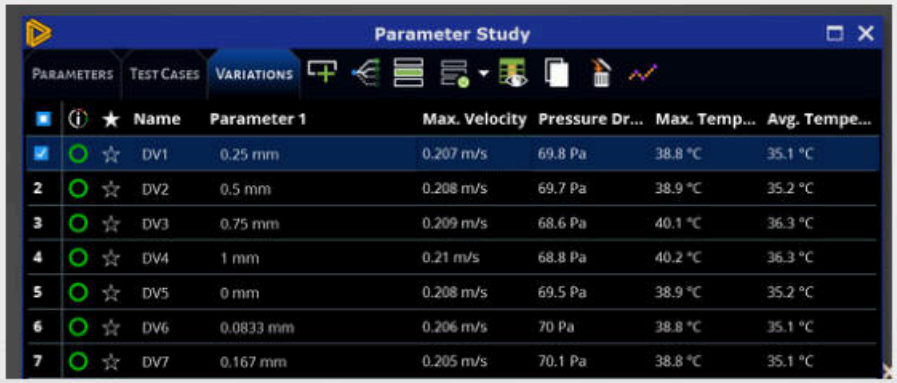
Summary

Overall, the velocity distribution allows heat to be removed efficiently along the channels, but slight variations in heat flux between channels result in a temperature profile that is low but not perfectly uniform.

Optimisation





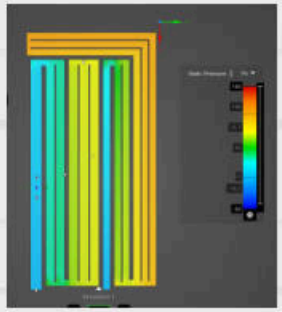


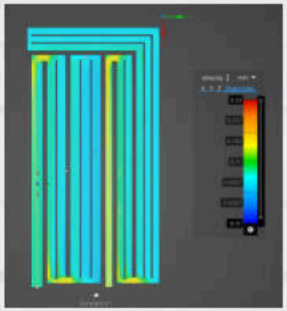
Width adjustment：

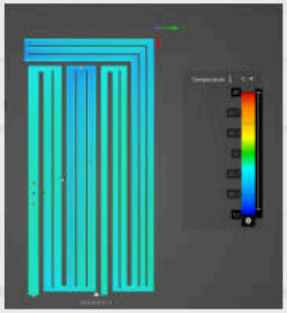
A parametric study was conducted to investigate the effect of reducing channel width on coolant flow behaviour and thermal performance. Multiple design variations were tested to evaluate how channel width influences velocity, pressure drop, and temperature distribution.

Reducing the channel width increases coolant velocity due to the smaller flow area, which enhances convective heat transfer and improves heat removal from the cooling plate. However, narrower channels also increase hydraulic resistance, which can lead to a higher pressure drop.

Based on the parametric study results, the optimal configuration was obtained by reducing the channel width by 0.25 mm on both sides. This adjustment increases coolant velocity and improves cooling effectiveness while keeping the pressure drop within an acceptable range, resulting in a better balance between thermal performance and hydraulic efficiency.



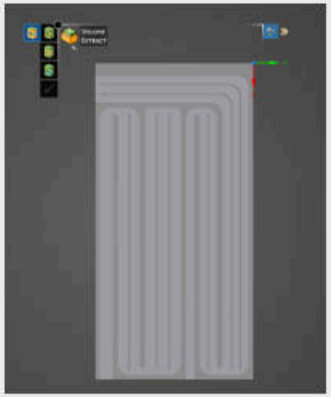


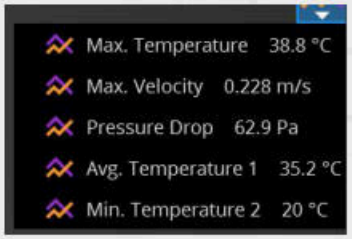


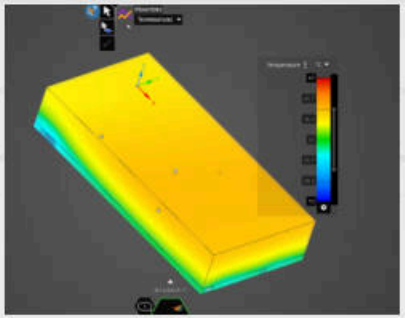
Smoothen the pass transitions and junctions

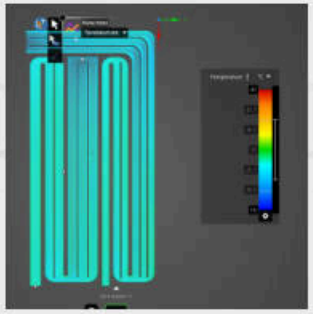
To further improve the flow behaviour, the transitions and junctions between flow paths were smoothed to reduce sharp corners and abrupt geometric changes. Such geometric discontinuities can cause additional pressure losses and disturb coolant distribution.

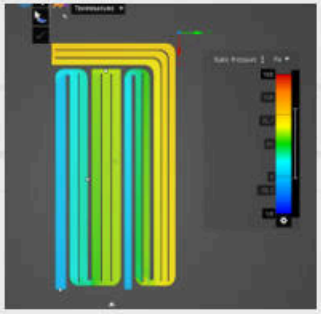
After smoothing these transitions, the coolant flow becomes more stable and evenly distributed across the cooling channels. The CFD results show that the pressure drop is reduced to approximately 62.9 Pa while maintaining similar temperature performance, indicating improved hydraulic efficiency without compromising cooling capability.

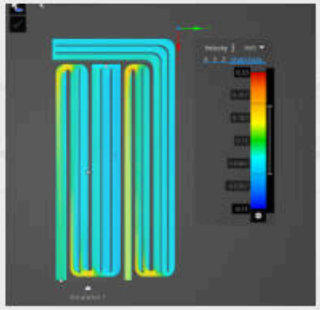










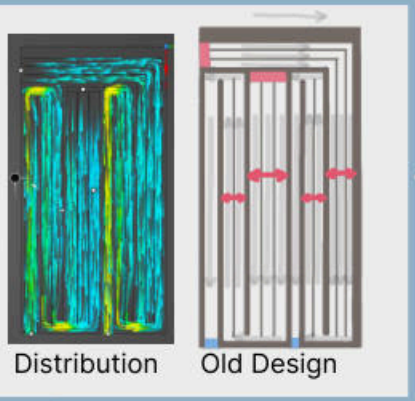


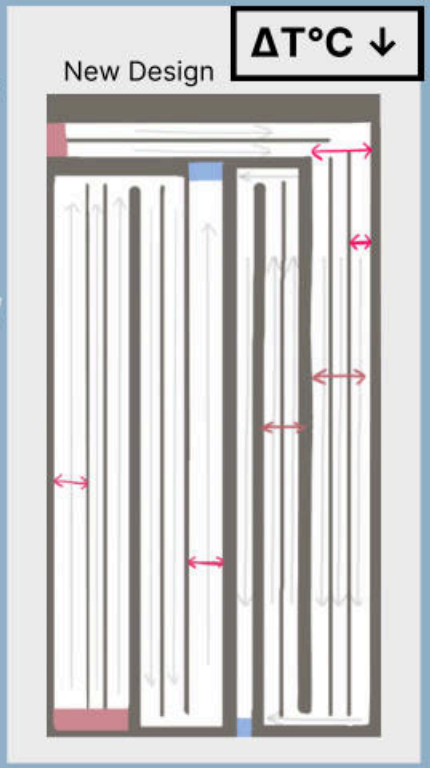
Future Recommendations

After analysis, comparing and optimising our results, we are able to scrutinise the design and use that to suggest possible redesign solutions.

Wider Downstream

Issue:
Flow maldistribution highlighted by the CFD, particularly in the downstream and outer channels.





Solution:
The downstream or outer channels can be widened slightly or their path lengths shortened to equalise hydraulic resistance across the parallel network. This is evidenced by the results: thermal spread in the battery is high at 13.5°C and the pressure drop is low at 69.5 Pa. There is hydraulic headroom for decreasing this thermal spread by widening/shortening.

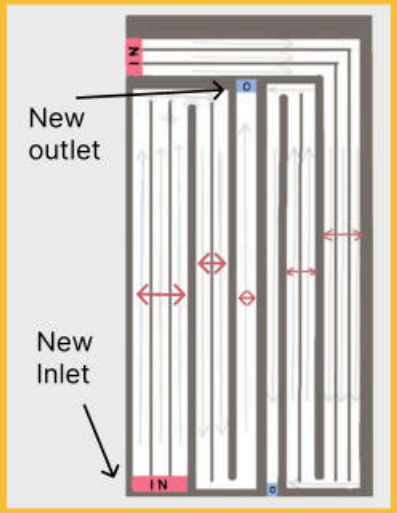
Part of the design rationale was to vary channel lengths to maintain even heat transfer; this recommendation takes it a step forward, not only are we controlling the velocity, we are also correcting the uneven resistance caused by the outside/ downstream channels.

Expected Outcome:

- Reduce velocity differences between channels and improve coolant delivery to the warmer downstream channels.

Counter Flow

Issue:
The heat flux is significantly higher at the two inlets, which are positioned close to each other, and the cooling effectiveness decreases, causing uneven cooling.

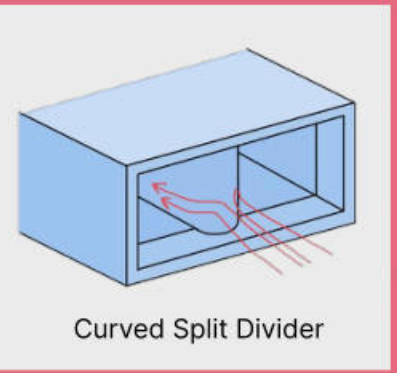


Solution:
Reconfigure the the two paths in a counterflow arrangement so the loops flow in opposite directions, which will flatten out the overall temperature profile across the plate. This arrangement addresses the weakness, by placing warm and cold coolant next to each other. This builds on the MPS design, it uses the controlled routing advantage of a multi-pass design to improve the thermal driving-force distribution across the plate.

Expected Outcome:
This will improve thermal uniformity over reducing maximum battery temperatures. The benefit is a reduction of downstream hot regions.

Smoothed Junctions and transitions

Issue:
There are many regions (inlet, splits, bends, merges) that currently cause local pressure losses and reduce the flow stability, which effects the flow distribution.



Solution:
Further smoothing of all the junctions and transitions will decrease the pressure losses, as suggested during optimisation. Angled bifurcations will allow for more gradual transitions between flow regions.
These losses come from: flow separation at corners, small vortices forming at splits and merges, and sudden acceleration/deceleration of the fluid.

Expected Outcome: This will improve flow uniformity, leading to more consistent cooling, lower peak temperatures and a more uniform temperature spread.

Appendix

References

Car

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Battery

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[7]Heliyon (2024) Research article. Available at: https://www.sciencedirect.com/science/article/pii/S2352152X24042944 (Accessed: 28 February 2026)

Team Roles

Pages

Charlie- 2, 7, 8, 9, 16, 18
Yousuf- 4, 5, 6, 16, 18
Zayn- 2, 3, 6, 15, 18, 19
Zipei-6, 7, 10, 11, 12, 13, 14, 17

Content

Charlie- Hand Calculations, Battery Analysis
Yousuf- Car, Battery Analysis
Zayn- Car, CFD and Calculation Comparison
Zipei- CFD Setup, Simulation, Results and Optimisation

Project Management

Held meetings at least once a week speaking about:

- Lectures and Content
- Individual Research and Insights
- Reviewing work done throughout the week
- Split work up
- Set deadlines throughout the week before next meeting

